

RUSHA BHAVESH MISTRY

Long Beach, CA · +1 (562) 341-6389 · rusha406@gmail.com · linkedin.com/in/rushamistry

SUMMARY

Software engineer with 1+ year of production experience building Python backends, REST APIs, and AI-powered applications. Skilled in large language model (LLM) systems, retrieval-augmented generation (RAG), and machine learning, with a focus on shipping reliable, scalable, and well-tested software. Pursuing an M.S. in Computer Science (GPA 3.7) with an emphasis on Advanced AI, Machine Learning, Algorithms, and Databases.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript/TypeScript, HTML/CSS

AI / ML / LLM: LLM application development, Retrieval-Augmented Generation (RAG), prompt engineering, embeddings, semantic search, NLP, transformers, TensorFlow, Hugging Face, scikit-learn, OpenAI API, LangChain

Backend / APIs: FastAPI, Flask, REST API design, AsyncIO, event-driven architecture

Data / Databases: PostgreSQL (pgvector), MySQL, MongoDB, vector databases, data pipelines, Pandas, NumPy

Cloud / DevOps / Tools: AWS, Azure, Docker, CI/CD (GitHub Actions), Git, pytest, Postman, RAGAS

EXPERIENCE

Python Engineer

Dec 2023 – Dec 2024

PayGURU Technosoft Private Limited

Vadodara, India

- Built and maintained 10+ Python backend services and REST APIs (FastAPI) across 25+ client sites, cutting manual operations by 60%+ via automated onboarding, sync, and reporting.
- Designed ETL and data pipelines processing 50,000+ daily records into validated datasets, with monitoring and alerting that sustained 99.5%+ uptime in production.
- Developed event-driven automation services and scheduled jobs in Python to streamline repetitive operational workflows.
- Triageed and resolved 100+ defects via root-cause analysis and hotfix deployments for 5,000+ active users.

Data Science Intern

Sept 2023 – Jan 2024

SlashMark IT

Remote

- Developed Python and SQL backend APIs and data workflows supporting 2,000+ users, taking 3+ features end-to-end from scoping to deployment.
- Built data processing and image classification modules with Python ML libraries, reducing manual review effort by approximately 35%.
- Refactored legacy services and added unit tests to eliminate recurring production failures.

PROJECTS

LLM-Powered Knowledge Assistant

Python, FastAPI, LangChain, OpenAI API, pgvector, Docker

- Built an LLM-powered retrieval-augmented generation (RAG) application that answers natural-language questions over a large knowledge base using semantic search and source-grounded generation.
- Engineered the retrieval pipeline – chunking, embeddings, and vector search over pgvector with reranking – to improve relevance and minimize hallucinations.
- Added a RAGAS-style evaluation harness to catch quality regressions and flag low-confidence responses before output.
- Containerized the service with Docker.

Real-Time Risk Detection & Alerting Pipeline

Python, TensorFlow, OpenCV, AsyncIO, REST APIs

- Trained a CNN image classifier and fused predictions with live weather APIs to generate context-aware risk alerts.
- Cut end-to-end inference latency by 50%+ using AsyncIO-based concurrent fetching and batched TensorFlow inference.
- Structured modular, independently testable components (ingestion, validation, inference, alerting) for reliable deployment.

NLP Sentiment Analysis & Triage System

Python, Hugging Face, BERT, NLTK, SQL

- Built a dual-model NLP pipeline (lexicon scoring plus fine-tuned BERT) for sentiment classification, benchmarking accuracy trade-offs and persisting results to a normalized SQL schema.

EDUCATION

M.S. Computer Science, California State University, Long Beach (CSULB)

Long Beach, CA · Expected 2027

GPA: 3.7 / 4.0 · Coursework: Advanced AI, Machine Learning, NLP, Algorithms, Database Systems, Distributed Systems

B.Tech, Computer Science & Engineering, Parul Institute of Engineering & Technology

Vadodara, India · 2024

GPA: 8.04 / 10

CERTIFICATIONS & ACHIEVEMENTS

Microsoft **AZ-900** (Azure Fundamentals) · Microsoft **PL-900** (Power Platform) · **Machine Learning Specialization** (Coursera)

· Google Data Analytics (Coursera) · **Winner**, Pseudo Heist regional tech fest